

SHAP Integration with the DeBERTa-v3-large Gated-Hybrid Model

1. Introduction

Email phishing detection is a high stakes text-classification problem because apparently persuasive messages may exploit lexical cues, contextual framing and structural properties simultaneously. Transformer encoders offer powerful contextual representations, but their predictive capacity does not itself establish why an individual email was classified as phishing. A gated-hybrid design further complicates interpretation: a model may combine a learned contextual vector with explicit engineered features, with relative contributions conditioned on the input. This review examines whether and how SHapley Additive exPlanations (SHAP) can support defensible local interpretation of such a system. The central argument is that SHAP offers a coherent attribution framework, but explanatory validity depends critically on the model interface, feature grouping, background distribution, and output scale (Lundberg and Lee 2017).

2. Background and key concepts

DeBERTa introduced disentangled attention representations of content and relative position (He et al. 2021). DeBERTaV3 subsequently adopted replaced token detection and gradient disentangled embedding sharing to strengthen pre-training efficiency and natural language understanding performance (He, Gao and Chen 2023). Neither study directly validates phishing detection or the interpretability of a project-specific gated-fusion architecture. Transfer to email classification therefore requires task-specific evaluation.

In the project architecture, the subject and body form model text for the DeBERTa encoder; a parallel branch computes twelve engineered email features; a learned gate combines contextual and engineered representations before binary classification. A gate weight is an internal mixing parameter, not automatically a faithful feature attribution. Correlated input features, interactions and non-linear transformations mean that neither a gate coefficient nor an attention weight can be read as a standalone causal explanation (Jain and Wallace 2019).

SHAP explains a chosen scalar model output relative to an expected baseline by allocating additive contributions among a defined set of players or features. For an explanation expressed in the same output space as its baseline, baseline plus feature contributions approximates or equals the explained model output, subject to explainer assumptions and approximation. SHAP values are not intrinsically probabilities: they may be computed on logits, log-odds, probabilities or another model output. The sign of an attribution must therefore be interpreted relative to the target class and output being explained (Lundberg and Lee 2017).

In hybrid NLP, a crucial design choice is what counts as a feature. Possibilities include text tokens, masked text spans, the twelve engineered variables, or grouped input modalities. If SHAP is applied only to the engineered-feature branch, the resulting explanation does not establish token-level influence from DeBERTa. Conversely, token-

only explanations may omit the contribution of engineered features. A full-model explanation requires an interface that recomputes the actual final prediction under the chosen perturbations, preserving the intended relationship between textual and derived features.

3. Review of existing research

Lundberg and Lee (2017) formulate SHAP as a unified additive feature attribution framework drawing on Shapley values. Its conceptual contribution is particularly useful for multi source classifiers. Attributions can be stated against a common baseline rather than relying on incompatible raw feature scales. However, the theoretical properties depend on the value function and feature coalition semantics. In language applications, replacing words or masking spans can produce unnatural inputs, so mathematically well-defined allocations are not necessarily faithful to realistic email edits.

He et al. (2021) establishes the representational foundations of DeBERTa, while He, Gao and Chen (2023) document improvements in DeBERTaV3 pre-training. These papers support selecting a strong contextual encoder, but do not demonstrate that the encoder's attention distribution explains phishing decisions. Jain and Wallace (2019) found that attention weights can diverge from other importance measures and that alternative attention distributions can yield comparable outputs. Wiegrefe and Pinter (2019) qualified the broad rejection of attention explanations, arguing that their assessment depends on the explanatory claim and testing protocol. Taken together, these studies justify testing explanations empirically instead of equating visual salience with model reasoning.

Ribeiro, Singh and Guestrin (2016) propose LIME, which approximates a classifier locally using an interpretable surrogate. LIME can provide accessible word-level rationales but is sensitive to the perturbation distribution and locality specification. Sundararajan, Taly and Yan (2017) propose Integrated Gradients, attributing predictions along a path from a baseline input; it is computationally attractive for differentiable networks but depends on the baseline and gradient-accessible representation. These approaches are complementary comparison points rather than proof that SHAP is universally superior.

Adebayo et al. (2018) show why visually convincing saliency maps cannot be accepted without model- and data-sensitivity checks. Their experiments concern saliency methods and do not directly prove SHAP failure on phishing data; nevertheless, their evaluation principle generalises: an explanation should respond meaningfully to relevant changes in model parameters and inputs. For a gated-hybrid classifier, this motivates checking whether explanations change under text perturbation, feature ablation and model variation.

4. Comparison of existing methods or solutions

Approach	Interpretive target	Strength	Critical limitation
SHAP	Local contribution to specified output	Common additive framing across feature groups	Cost; masking/background dependence; correlated features
LIME	Local surrogate decision boundary	Model-agnostic, accessible local rationale	Perturbation and sampling instability
Integrated Gradients	Gradient-path attribution	Uses differentiable model structure	Baseline/path dependence; multimodal input complexity
Attention visualization	Internal attention distribution	Available inside transformer layers	Not equivalent to prediction attribution
Gate inspection	Learned mixing behavior	Describes fusion mechanism	Gate values alone do not quantify causal or SHAP contributions

No method resolves all interpretability challenges. SHAP is well suited to explaining an explicitly specified final prediction, while attention and gate inspection may provide complementary diagnostics. The project should avoid presenting these distinct quantities as interchangeable evidence.

5. Research gaps and limitations

First, the literature on pretrained transformers does not establish explanation fidelity for a specific DeBERTa-v3-large gated-hybrid phishing classifier. The combination of contextual text and engineered features create cross branch interactions that may be obscured by branch specific attributions. Second, feature dependence matters: urgency words, URL counts and suspicious link indicators may co-occur and independently masking them can generate implausible emails. Third, explanation results are conditional on the background distribution and masking policy; changing these may alter the magnitude and rank of reported factors.

Fourth, the computational burden of explaining a large transformer can constrain interactive latency. Sampling based explainers may trade runtime for precision, and explanations generated on a surrogate or reduced input are not automatically equivalent to explanations of the full classifier. Fifth, displayed “supporting” and “opposing” factors are only valid relative to the output class and SHAP sign convention actually implemented. Ranking by absolute SHAP value identifies magnitude, not whether a factor favours phishing or safe classification. Finally, user-facing explanations risk overstating causal certainty: SHAP is a local attribution of model behaviour, not proof that a suspicious phrase caused a malicious intent.

6. Implications for the proposed project

The project should define a single documented explanation target: for example, the final model phishing class logit or phishing probability. It should state which output is explained, how the baseline is selected, what the perturbations change and whether the twelve engineered features are recomputed from perturbed text or treated as separate players. Where the explainer operates on only one branch, the interface should label that scope explicitly rather than claiming a full-model explanation.

The user interface can group signed attributions into “Factors Supporting the Prediction” and “Factors Working Against the Prediction”, with “Strongest SHAP Factors” ranked by absolute contribution. The explanation should identify the class being explained, include the baseline/output scale where feasible and distinguish detected email indicators from SHAP derived contributions. For safe predictions, the direction of supporting factors must be recalculated or reinterpreted consistently with the safe class output rather than reusing phishing class signs uncritically.

A defensible evaluation plan would compare attribution rankings under repeated runs, alternative backgrounds and semantically constrained email edits, test whether removing highly attributed evidence affects the model score more than removing low attribution evidence and inspect disagreement between SHAP and other attribution methods. These are recommended validation experiments, not results already established by the current project. The same model prediction should be explained consistently across Streamlit and the browser extension, although rendering differences may be acceptable. Response latency should also be measured separately for classification and explanation.

Finally, reporting should link example explanations to the actual test inputs, predicted labels, confidence values, feature names and explainer configuration. This audit trail is essential to distinguish a plausible demonstration from a reproducible explanation pipeline.

7. Conclusion

Research supports using DeBERTaV3 as a capable contextual encoder and SHAP as a principled local attribution framework, but does not justify treating the resulting hybrid system as inherently interpretable. In a gated-hybrid phishing classifier, interpretability depends on the boundary of the explained model, coalition definitions, dependence assumptions and validation against the actual prediction function. The strongest project contribution is therefore not merely displaying SHAP charts: it is implementing and clearly qualifying explanations of an operational email classifier, then evaluating their fidelity, stability and usefulness across interfaces.

References

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